CYBERSENTINEL COPILOT

Full project flow and implemented architecture

CIC-IDS2017 · SQLite + FTS5 · local ML · Synapt MCP · FastAPI · LangGraph · human review

A · OFFLINE FOUNDATION AND RUNTIME ENTRY

CIC-IDS2017 preparation

8 Parquet files · 78 source columns
Audit, normalize, label-map
Preserve file + row provenance
2,313,810 flows

SQLite evidence layer

flows table · exact filters
291 incident summaries
FTS5 BM25 keyword index
analyst_feedback table

Local prediction models

65 traffic features · 9 families
HistGradientBoosting classifier
IsolationForest anomaly detector
Versioned joblib bundles

Synapt knowledge layer

Curated MITRE + NIST context
Historical semantic retrieval
TLS/auth/JSON-RPC adapter
Knowledge, RAG, judge roles

Analyst dashboard

Investigate · search · chat
Review evidence and limitations
Submit feedback
HTML / CSS / JavaScript

FastAPI boundary

Pydantic validation
Nine HTTP routes
JSON response shaping
Shared service instances

flow + BM25

model inference

semantic + guidance

B · POST /investigate/flow — SEVEN-STAGE LANGGRAPH CHOREOGRAPHY

1 · Guardrail

Pydantic + query checks
Injection/action patterns
Domain relevance

2 · Flow triage

Record lookup / 404
FTS5 + semantic RAG
Self-exclusion

3 · ML analysis

Family + confidence
Risk + top deviations
Anomaly score

4 · Knowledge

Sanitized Synapt query
MITRE/NIST context
Local fallback

5 · Priority

Deterministic P1–P4
Family + risk + anomaly
Missing context

6 · Response

Investigation steps
Recommendations
Provenance + limits

7 · Judge

Hard safety gate
Four rubric dimensions
Provider/fallback scores

C · RETURNED EVIDENCE AND HUMAN DECISION

Structured response to dashboard

Attack family · confidence · intrusion risk · severity · anomaly · P1–P4 priority · feature deviations · similar incidents
Synapt analysis · MITRE mappings · investigation steps · response drafts · judge verdict · source provenance · limitations

Human analyst decision

Verify evidence · approve actions
No automated containment

D · SUPPORTING API, FEEDBACK, AND EVALUATION PATHS

Search · /search/incidents

Structured SQLite filters
FTS5 BM25 + Synapt channel
RRF-ranked incident summaries

Copilot · /chat

Guardrails + optional record
Classifier context + MITRE
Synapt or local response

Feedback lifecycle

Submit → pending review
Human accept/reject gate
Holdout + candidate report

Evaluation evidence

40 structured fixtures
DeepEval adapter + JSON report
Model/anomaly benchmarks

Health + agent status

SQLite live readiness
Model/RAG indicators
Synapt configuration status

IMPLEMENTATION BOUNDARIES FOUND DURING SOURCE REVIEW

Dataset metadata

No IPs, ports, Flow ID, or timestamps
in this local CIC-IDS2017 export.

Retrieval

Semantic fusion needs provider correlation;
short result sets use random padding.

Evaluation + judge

DeepEval case score is fixed locally; judge
uses fixed defaults when provider fails.

Improvement + health

Candidate copies active model; health reports
Synapt configuration, not a live probe.

runtime flow

offline/shared dependency

Source-checked against the current workspace · generated by scripts/render_architecture.py